

KONSTANZ, GermanyWMO: **109290**Lat: **47.6775N**Lon: **9.1900E**Elev: **444**StdP: **96.10**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.3	-13.0	1.3	-9.4	-10.3	1.6	-5.9	10.2	6.4	8.7	5.7	2.1	0	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	29.7	19.6	27.9	18.9	26.2	18.2	20.6	27.3	19.7	25.8	19.0	24.7	2.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.9	23.1	17.6	13.3	22.4	16.9	12.7	21.7	61.3	27.6	58.3	25.9	55.7	24.8	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.5	6.2	5.2		-10.9	32.6	4.2	1.4	-14.0	33.6	-16.4	34.5	-18.8	35.3	-21.9	36.3	
			DB	-11.4	22.3	4.1	0.7	-14.3	22.8	-16.7	23.3	-19.0	23.7	-21.9	24.2	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	0.3	0.9	5.4	8.8	13.7	17.1	19.8	19.1	15.6	10.5	4.5	1.8
	DBStd	7.78	4.57	4.15	3.64	3.40	3.80	3.74	3.10	3.09	3.07	3.29	3.51	3.61
	HDD10.0	1236	301	256	149	62	13	1	0	0	1	33	167	253
	HDD18.3	3259	559	489	402	286	148	67	21	28	91	242	414	511
	CDD10.0	1181	0	0	5	26	127	213	304	283	169	51	2	1
	CDD18.3	163	0	0	0	0	3	30	67	54	9	0	0	0
	CDH23.3	1216	0	0	0	2	42	235	524	363	50	1	0	0
	CDH26.7	293	0	0	0	0	2	55	144	88	3	0	0	0
Wind	WSAvg	1.8	2.1	2.1	2.2	2.1	1.8	1.8	1.6	1.5	1.5	1.6	1.8	1.9
Precipitation	PrecAvg	836	45	46	53	61	88	102	99	92	70	62	59	67
	PrecMax	1046	134	113	146	138	185	174	202	192	159	115	145	159
	PrecMin	617	2	14	16	7	15	31	26	14	32	9	1	3
	PrecStd	110	32	24	30	35	43	35	43	47	32	30	33	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	13.6	20.3	22.5	26.5	30.6	32.1	31.7	26.7	21.9	15.1	12.0
		MCWB	8.0	8.7	12.1	12.7	16.0	19.3	20.4	20.0	18.4	15.8	10.6	8.8
	2%	DB	9.4	10.2	15.3	19.7	24.4	28.2	30.1	29.1	24.8	18.6	12.2	10.4
		MCWB	6.8	6.8	9.1	11.3	15.2	18.8	19.8	19.4	18.0	14.4	9.9	8.2
	5%	DB	7.7	8.0	12.8	17.5	22.8	26.0	28.2	27.1	22.9	16.6	10.6	8.7
		MCWB	5.6	5.7	7.8	10.4	14.4	17.9	19.1	18.8	17.1	13.1	8.8	6.9
	10%	DB	6.0	6.3	10.7	15.2	20.7	24.0	26.3	25.2	21.0	14.8	9.1	6.7
		MCWB	4.5	4.6	7.0	9.4	13.5	16.9	18.6	18.1	16.1	12.3	7.6	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	9.0	11.8	13.4	17.3	21.1	21.8	21.4	19.5	16.3	11.6	10.0
		MCDB	11.0	11.9	19.4	20.9	24.1	27.2	29.8	28.9	24.9	20.5	13.6	11.8
	2%	WB	7.1	7.3	9.8	11.9	15.9	19.8	20.8	20.2	18.4	15.0	10.2	8.2
		MCDB	9.1	9.5	14.2	18.5	22.6	25.4	27.9	26.8	23.6	17.5	11.9	9.8
	5%	WB	5.8	6.0	8.5	10.9	15.1	18.7	19.9	19.4	17.6	13.7	9.0	6.9
		MCDB	7.4	8.0	11.8	15.9	21.0	24.1	26.3	25.3	21.8	15.7	10.6	8.5
	10%	WB	4.6	4.7	7.5	10.0	14.2	17.6	19.1	18.7	16.6	12.6	7.8	5.4
		MCDB	5.9	6.2	10.1	14.6	19.3	22.9	25.1	24.0	20.0	14.4	9.0	6.7
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.7	7.9	9.2	9.9	9.6	10.3	9.9	8.7	6.5	4.7	4.0
		MCDBR	6.6	8.3	11.5	13.9	13.7	12.9	13.4	13.0	11.2	8.8	6.7	6.3
		MCWBR	4.4	5.5	6.2	6.9	6.3	5.2	5.6	5.2	5.4	4.8	4.7	4.6
	5% WB	MCDBR	6.3	7.8	10.1	12.1	11.9	12.0	12.0	11.7	10.5	8.4	6.3	6.0
		MCWBR	4.6	5.4	5.8	6.2	5.7	4.9	5.3	5.0	5.2	4.9	4.9	4.7
Clear-Sky Solar Irradiance	taub	0.293	0.317	0.352	0.389	0.401	0.407	0.409	0.402	0.376	0.347	0.316	0.288	
	taud	2.443	2.377	2.349	2.278	2.284	2.305	2.293	2.330	2.382	2.451	2.486	2.451	
	Ebn at Noon	794	843	864	862	863	858	849	840	829	799	758	753	
	Edh at Noon	65	87	104	123	128	126	126	116	100	80	63	58	
All-Sky Solar Radiation	RadAvg	1.11	1.93	3.13	4.31	5.00	5.73	5.50	4.80	3.59	2.24	1.23	0.90	
	RadStd	0.12	0.27	0.42	0.61	0.51	0.52	0.50	0.49	0.37	0.24	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (111 neighbors)	N/A	N/A	N/A	N/A	+0.43	+0.37	N/A	N/A	+54	+14

Nomenclature: See separate page